

1. Academic Core and Content

Entity	Attributes	Description
Course_Unit	ID_UC(PK) , ID_Professor(FK), Name, Semester, Curricular_Year	Ex: Digital Systems (SDIG), Computer Architecture (AC).
Topic	ID_UC (FK) , Name (PK) , N_Order	Ex: Logic Gates, Karnaugh Maps, MIPS Instruction Set.
Teaching_Material	ID_Material (PK) (UUID) , ID_UC (FK), ID_Professor (FK), Status (Pending, Indexed, Error), Title, File_Path, Extracted_Text, Upload_Date	PDF uploaded by the professor to serve as a base (RAG) for the LLM.
Exercise	ID_Exercise (PK) (UUID) , ID_UC(FK), Topic_Name (FK), Material_Ref (FK), Type, Question, Solution (JSONB to support different types of questions), Difficulty, Explanation	Can be AI-generated or fixed. Types: Multiple Choice, Truth Table, K-Map.

2. Users and Personas

Entity	Attributes	Description
Base_User	ID_User (PK) (UUID) , Name, Email, Password_Hash, Role, Registration_Date	Central table containing credentials and data common to all system users.

Student	ID_Student (PK/FK), Current_Level, Total_XP, Streak_Days, Last_Access	Contains gamification and progress data. ID_Student directly references the ID_User from the base table.
Professor	ID_Professor (PK/FK), Department, Office, Short_Bio	Professional data of the teacher. ID_Professor references ID_User. Allows linking the teacher to Course Units and Materials.
Professor_UC	ID_Professor (PK/FK), ID_UC (PK/FK)	Tabela de junção que estabelece quais professores têm permissão para gerir uma determinada UC
Admin	ID_Admin (PK/FK), Privilege_Level (1-3), Contact	System management data. ID_Admin references ID_User. Responsible for approving Requests.

3. Gamification and Progression

Entity	Attributes	Description
Progress	ID_Progress (PK - UUID) , ID_Student (FK), ID_Exercise (FK), Attempts, Status, Date, XP_Earned, Sync_Status	Records if the exercise was solved correctly and the reward obtained (XP).
Streak	ID_Streak (PK - UUID) , ID_Student (FK), Log_Date (Date), Sync_Status	To validate if the user studied consecutive days (crucial for retention).

4. Admin Related

Entity	Attributes	Description
Request	ID_Request (PK) , ID_Professor(FK), ID_Admin (FK), Title, Description, Status (pending, approved, rejected), AdminComment, Creation_Date, Resolution_Date	The Professor sends requests to admins.
Admin_Audit_Log	ID_Log (PK) , ID_Admin (FK), Action, Target_ID, Target_Type, Date	Security: Records who changed permissions or deleted Course Units. Critical security requirement. Note: TargetType can be Student, UC, etc.

Analysis

Student (End User)

- Consumes: Specific content from their Course Unit
- For each correctly answered exercise, student progress increases, i.e., generates N records of Progress when answering exercises.
- Has 1 active Streak record validated daily.
- Consumes exercises from Course Path

Professor

- Manages: 1 or more Course Units (UCs).
- Owns: 0 or more Teaching Materials (PDFs/Slides).
- Adds Students.
- Creates and Moderates: Exercises
- A professor can create 0 or more exercises.
- Petitions: 0 or more Action Requests to the Administrator.

Admin

- Creates Course Units.
- Associates Professors with courses.
- Can have 0 or more unanswered requests from Professors.
- An Admin can delete N users.

Course Unit

- Contains 1 or more Topics.
- Belongs to 1 or more Professors.
- Groups Teaching Materials that serve as a base for the RAG.

Exercise

- Mandatory linked to 1 Topic and 1 Course Unit.
- If AI-generated, must reference the Teaching Material that originated it.

Topic

- A topic has 1 or more exercises.

Teaching Material

- Each file is uploaded by 1 professor.
- All professors can upload Teaching Material files.